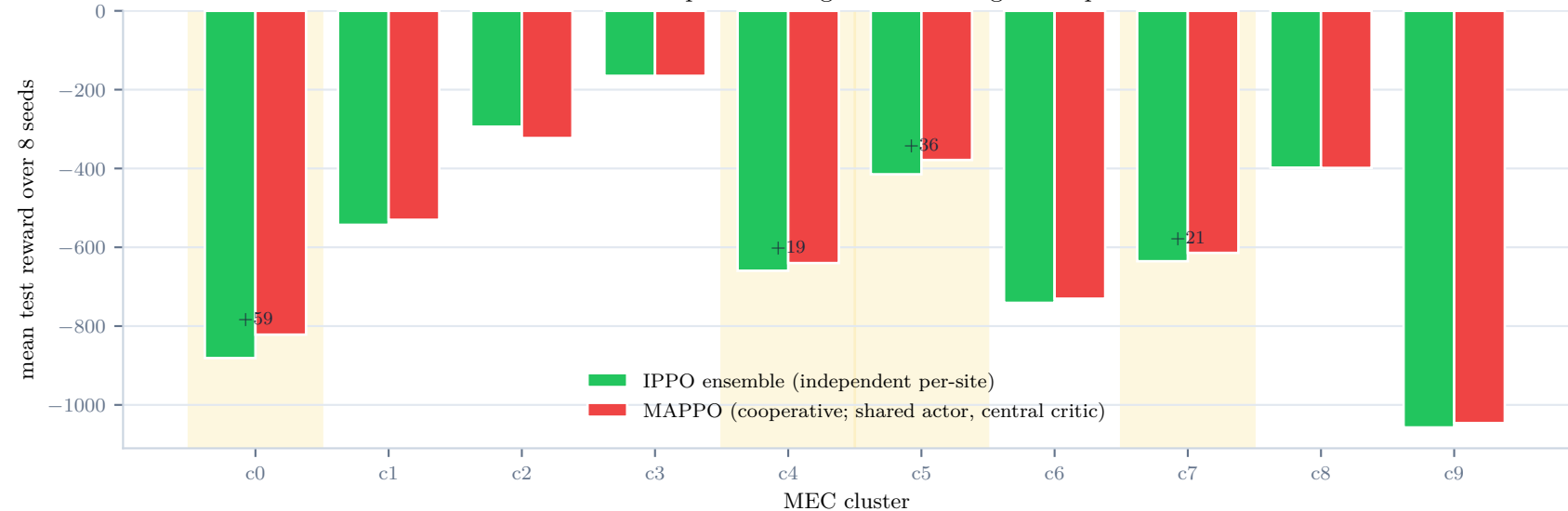


Per-cluster reward: independent single-site learning vs cooperative MAPPO



Shaded: MAPPO > ensemble by > 15 reward units — data-poor sites that inherit the fleet's operating rule.